Model-Base and Model-Free Control on Cart-Pole System

—a case study based on real-world interaction

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Abstract

Sim-to-real faces challenges such as the difficulty of building simulation environments or transfer errors. This paper explores the possibility of training and interacting with neural networks directly in real-world environments. The experimental subject is the Cart-Pole system. A hybrid approach combining Model-Based and Model-Free methods is used for control. Model-Based methods involve system identification using Neural ODE or SINDy, with segmented control based on energy methods and LQR. Model-Free methods use reinforcement learning algorithms for online learning and control policy generation.

Keywords

CART-POLE; MODEL-BASE; MODEL-FREE; DATA-DRIVE; NNODE; SINDY

1. Introduction

The development of machine learning and hardware has brought new transformations to control engineering. Neural network training relies on massive amounts of data, which requires extensive interaction with the environment. For safety and cost considerations, a common paradigm is sim-to-real—training neural networks in virtual environments and then transferring them to real-world environments for parameter fine-tuning. This approach presents two major issues: constructing virtual environments and transferring to the real world. The problem with virtual environments lies primarily in inaccurate descriptions of control objects, such as undefined friction or time-varying aerodynamics. The transfer problem mainly involves the difficulty of perceiving real-world parameters—for example, parameter feedback without latency in virtual environments does not exist in reality, which can lead to incorrect observations or a reduced observation domain for the neural network. These problems have been partially addressed by introducing random noise during training or using techniques like Nvidia's domain randomization. This paper explores the feasibility of training methods that operate directly in real-world environments.

2. Background

2.1. Cart-Pole System

The Cart-Pole is a classic control model (see fig1). It is often used as a benchmark for testing control methods. This system has a simple structure and, with proper constraints, does not risk damage from training failures. The system includes one stable equilibrium point and one unstable equilibrium point. Starting from the stable equilibrium, the environment exhibits self-recovery characteristics.

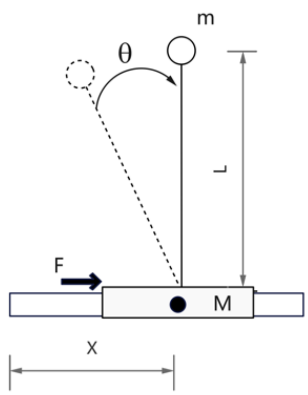

Fig1 Cart-Pole system

The dynamics of the Cart-Pole system can be described by four states $[x, \theta, \dot{x}, \dot{\theta}]$, and represented through the Lagrangian, which defines the relationship between kinetic and potential energy:

$$L(x,\theta,\dot{x},\dot{\theta}) = T(x,\theta,\dot{x},\dot{\theta}) - V(x,\theta) \tag{1}$$

via the Euler-Lagrange equations.

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial \dot{q}_i} = Q_i \qquad (i = 1,2,\ldots,n) \tag{2}$$

By applying the equations separately to the Cart and the Pole, the dynamic model of the system can be obtained:

$$(m + M)\ddot{x} - m\sin(\theta)\dot{\theta}^2 l + m\cos(\theta)\ddot{\theta} l = F \tag{3}$$

$$\ddot{x}\cos(\theta) + l\ddot{\theta} - g\sin(\theta) = 0 \tag{4}$$

2.2. Conventional Solutions

From the system's dynamic equations, it is evident that the system is nonlinear. However, near the unstable equilibrium point, the dynamics can be approximately linearized. Therefore, the typical approach is to divide the entire swing-up process into two tasks: Swing-Up and Balancing. The Swing-Up phase uses energy-based methods, such as Lyapunov energy derivative analysis. During the balancing phase, PID can be used for error control, or LQR can be applied for optimal control based on the system state.

2.2. Machine Learning Solutions

For basic control, reinforcement learning is commonly used. Many papers have discussed methods such as Q-learning, DQN, actor-critic policy gradient methods, and DDPG. While these methods differ in training efficiency, they have all been validated in virtual environments.

3. Building the Real-World Environment

Given that many studies use virtual platforms, I aim to build a physical experimental platform. Through direct interaction with real-world physics, corresponding control laws can be derived.

3.1. Architecture

The block diagram of the environmental architecture is as follows:

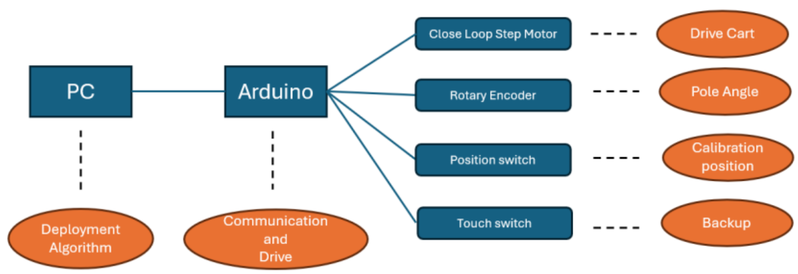

Fig 2 block diagram of the environment

The physical experimental platform uses a closed-loop stepper motor-controlled linear rail as the Cart component. A rotary encoder is connected to the inverted pendulum to form the Pole component. The Cart's position and velocity states are captured from the control signals of the stepper motor. The Pole's angle is obtained from the rotary encoder, and its angular velocity is calculated by differentiation.

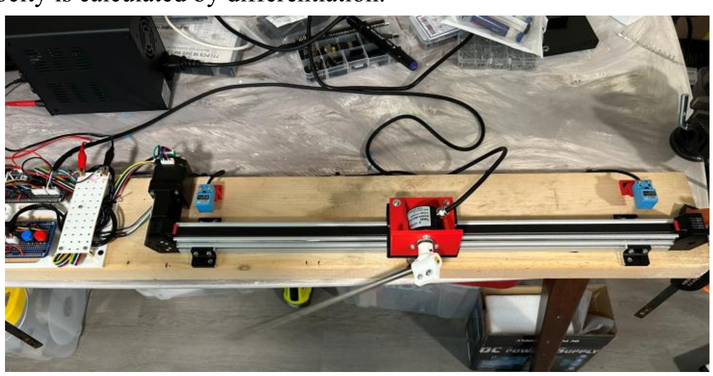

Fig 3 physical experimental platform

Unlike platforms that use DC motors, this platform employs a closed-loop stepper motor. The stepper motor allows direct specification of movement distance and speed through pulse count and pulse density, effectively changing the system input from force to position and velocity. This simplifies the system dynamics to:

$$\ddot{\theta} = \frac{g\sin(\theta) - \ddot{x}\cos(\theta)}{l} \tag{5}$$

3.2. Calibration

The experimental platform uses position sensors at both ends of the linear rail to detect the rail's limits, which are mapped to a custom standard position range of [-1.00, 1.00].

According to equation (5), when the Cart's acceleration is used as the input, the mass does not affect the system state. The length of the inverted pendulum should be automatically captured by the algorithm and applied in the law of control.

4. Deployment

4.1. Model-Free

Before applying the method directly to the physical platform, I tested the RSL_RL algorithm developed by ETH Zurich based on the Isaac Lab platform. This is a PPO algorithm with an actor-critic structure. Using 500 parallel Cart-Pole environments in Isaac Lab, training was completed in just 15,000 steps over 2 minutes.

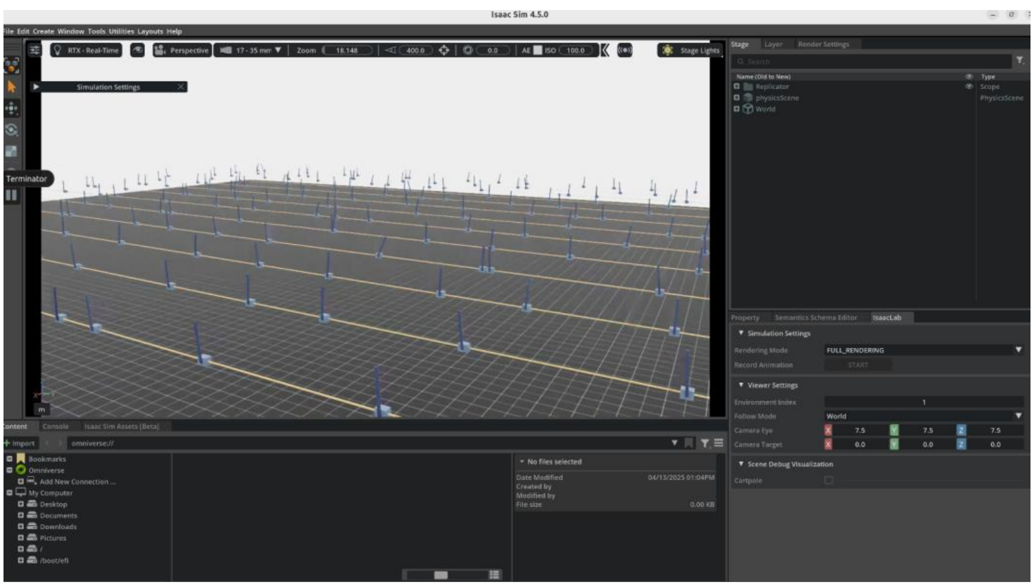

Fig 4 Isaac Lab Virtual Environment

Therefore, for the physical experimental platform, I also adopted the RSL_RL framework. The algorithm was deployed via a PC as the host computer. The observation space includes the Cart's position and velocity, as well as the Pole's angle and angular velocity. The output space is discrete positions [−0.02,0,0.02] with a constant velocity multiplier of 5. Both the actor and critic networks have two hidden layers with 128 units each.

In practice, the model discovered a swing-up strategy after 1,200 steps over 5 minutes, successfully swinging the Pole past the vertical unstable equilibrium point. However, it was unable to complete the subsequent balancing task.

4.2. Model-Base

The model-based control approach uses a sinusoidal input $X_{des}=A\cdot\sin(\omega t)$ to induce oscillations and accumulate energy for the swing-up. When the pole angle $\theta<10^\circ$, control is handed over to a PID controller, which stabilizes the system near the equilibrium point through error compensation. Due to its low computational cost, the PID controller is deployed directly on an Arduino Nano ESP32, eliminating the need for a host computer.

By tuning the frequency ω and amplitude A, the program is able to swing the inverted pendulum to the balancing position after 12 oscillation cycles. The PID controller can momentarily stabilize the pendulum for about one second but fails to maintain long-term stability.

5. A new approach

Through previous literatures* reading, we can conclude that both Model-Base and Model-Free control methods can solve the Cart-Pole task. Model-Free has a wider range of capabilities and can complete the entire task. The Model-Base method can only complete the Balancing stage task, and other methods need to be introduced for swing-up. However, Model-Base far exceeds the Data-Drive control method in terms of control speed and accuracy.

I conceived a hybrid control method. Combining the advantages of Model-Free and Model-Base. The overall idea is to use Model-Free as a real-time state sensor and Model-Base as a control strategy generator。The neural network generated by Model-Free should be learned online and updated online throughout its life cycle.

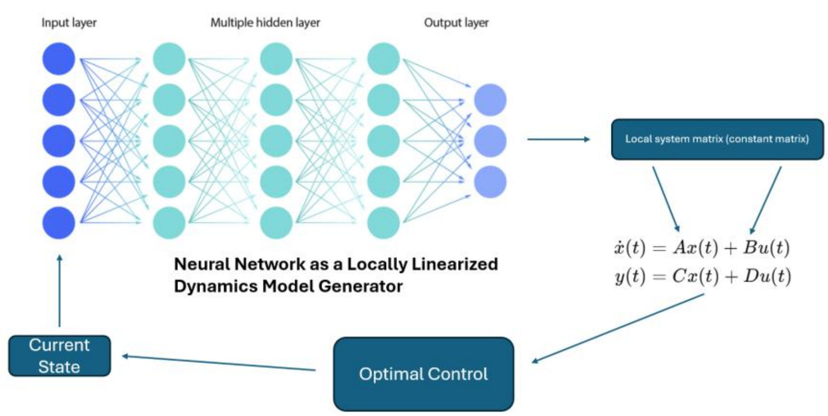

Fig 5 Neural Network State Generator

Here, an alternative approach is to adopt the DQN concept from reinforcement learning. DQN uses a trained neural network to generate real-time Q-values for policy-based action decisions, typically following a greedy or exploratory strategy. In this case, the training objective can be shifted from outputting Q-values to outputting the state-space at the current point. This state-space can be regarded as a locally linearized system model. Thus, the training process becomes a form of system identification. Unlike SINDy, which outputs a fitted nonlinear equation that must be linearized before applying linear control laws, the state-space output from the neural network can directly be used with a linear control law to compute the optimal action at the current point. This action drives the system into the next state, completing a control loop. This approach is somewhat similar to MPC, but with proper optimization, the neural network could offer an advantage in computational efficiency.

6. Conclusion

Although I began this project early and invested significant effort in building the hardware, unfortunately, I have not yet succeeded in completing it. Therefore, I can only submit this draft for now.

Below are the issues identified and potential improvements:

- Incorrect system input: The RSL_RL training used the wrong action space. According to equation (5), what truly affects the pole's angle state is the Cart's acceleration. The controller's driving function should be modified from position control to velocity or acceleration control.
- Low control frequency: The control frequency at which the Cart receives commands from the RSL_RL algorithm is far below 50 Hz, preventing the system from responding quickly to state changes. It remains unclear whether this limitation is due to the host computer's computation speed or the communication rate with the controller.
- Inaccurate state input: The system currently directly measures the Cart's position and the pole's angle. Velocities are calculated through differentiation, introducing errors. Moreover, calculated accelerations and angular accelerations are even less reliable. Future improvements should include low pass filtering or integration of acceleration sensors.

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